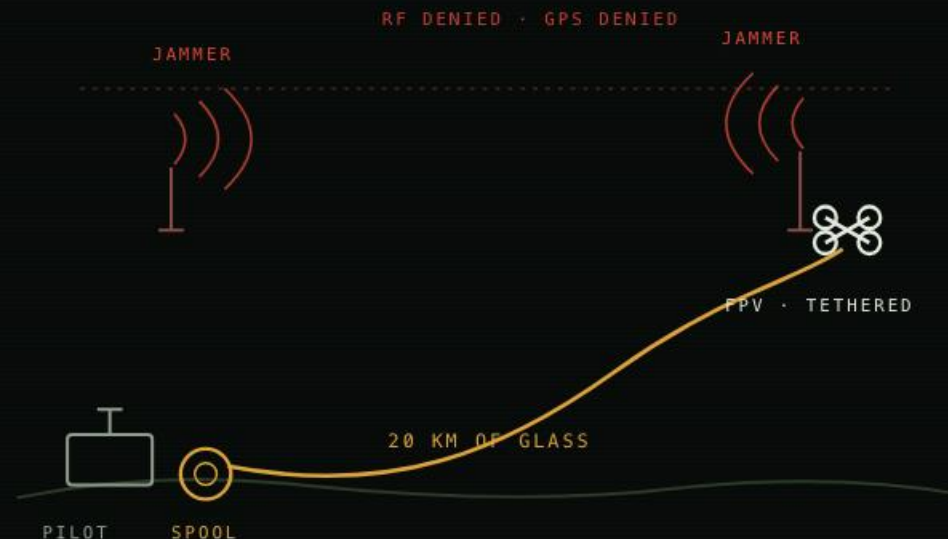


— VESPER · DN HACKS

The drone war is a comms war.

And right now, the state of the art is **a leash.**

EASHAN VAGISH · THEO OLTEAN



UKRAINE, TODAY: CREWS FLY STRIKE DRONES ON 20 KILOMETERS OF UNSPOOLING GLASS — BECAUSE THE SPECTRUM IS TOO JAMMED FOR RADIO. ORIGINAL ILLUSTRATION.

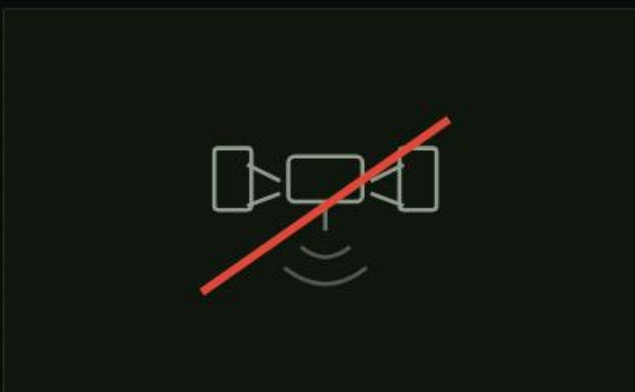
— THE PROBLEM

Out there, anything that phones home **dies.**



NO SPECTRUM

Key up a radio, become a target. The jammers own the air.



NO GPS

Every satellite fix is a lie. Fly by what you see — or don't fly.



NO EASY TARGETS

The armor isn't on anyone's map. It's under the trees, waiting.

And the line that never moves: **a human pulls the trigger. Every time.**

LIVE DOWNLINK

VESPER · TACTICAL

KRAMATORSK · AO 2000 M · N1

T+23.1s

POLICY · SEARCH.PT

RECENTER

SEARCHING

FOUND 1/3

NEUTRALIZED 0/3

ASSETS 16

LEAD SPD 9.4 m/s

AGL 86 m

VZ +4.2

TAKE CONTROL

W S fwd/back A D left/right R F up/down

policy search.pt

AO MAP

site frame · N up

KRAMATORSK_MS · 2000 M × 2000 M · N UP

live t=23 s · found 1/3 · reached 0/3 · search.pt

TRACK WAYPOINTS SIGHTED REACHED UNFOUND

drag to pan · scroll to zoom · click a marker

JOBS ON THE BOX

PULL ARTIFACTS

warm session

running · 559 min · STOP

2025-09-05T16:31:22Z [33.407...8ms] [Warning] [isaacsim.core.simulation_manager.plugin] No adjacent s

2025-09-05T16:31:22Z [33.407...497...8ms] [Warning] [isaacsim.core.simulation_manager.plugin] No adjacent s

2025-09-05T16:31:22Z [33.407...183572/30

2025-09-05T16:31:22Z [33.407...183572/30

2025-09-05T16:31:22Z [33.407...183572/30

You can't scale a cord.
We cut it — with software.

Sixteen drones that hunt in radio silence — and everything they need fits on a laptop we could carry into the field. No datacenter. No datalink. No leash.

THE VESPER CONSOLE — LIVE SORTIE OVER REAL KRAMATORSK TERRAIN, RUNNING ON THIS MAC.

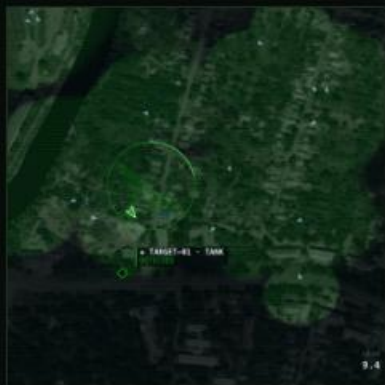
N

03 / 10

TO NAVIGATE

— THE HUNT

Find it. Call it in. Get the nod. Kill it.



01

SEARCH

One shared map. Sixteen hunters sweeping it.



02

DETECT

Armor in the treeline. Tagged in seconds.



03

UPLINK

Jammed? Hold the report. Fly to where the link lives.



04

APPROVE

The request hits your screen. You approve — or you don't.



05

STRIKE

One nod. One munition. One kill.

It hunts alone. It never kills alone.

— THE PIPELINE · 01 — RECONSTRUCT

We cloned the battlefield.

This background is not a photo — it's our sim. Type coordinates, and the pipeline reconstructs the real city in 3D from public satellite and terrain — every roof raised, every canopy grown, exactly where the real ones stand. Hours, not months.

48.7106°N 37.5762°E

01 · TYPE COORDINATES



02 · PULL SATELLITE + TERRAIN



03 · RECONSTRUCT IN 3D



04 · FLY IT — NATIVE SIM

KRAMATORSK — RECONSTRUCTED AUTOMATICALLY BY OUR PIPELINE. THE SWARM TRAINS, FLIES, AND DIES IN THIS WORLD.

— THE PIPELINE · 02 — TRAIN

A million sorties **before the first real one.**



LOST ———> METHODICAL ———> **LETHAL** · SAME POLICY, SAME TERRAIN — THE ONLY THING THAT CHANGED IS TRAINING

Millions of rollouts in the twin. Nobody wrote a waypoint. Nobody scripted a behavior. **Reward did the teaching — the kill lives in the weights.**

— THE PIPELINE • UNDER THE HOOD

Open data in. **A hunting swarm out.**

Every source is public. Every algorithm is ours. Running today — not a roadmap.

— DATA IN

ESRI WORLD IMAGERY • NAIP	orthophoto ground truth, down to 1 m
USGS 3DEP LIDAR • COPERNICUS GL0-30	elevation — 1 m US, 30 m anywhere on Earth
DRONE-SURVEY DSMs	OpenDroneMap photogrammetry, ingested as GeoTIFF
OPENSTREETMAP • OVERPASS	building footprints + heights, roads, rail, water, forest

— WHAT WE DO WITH IT

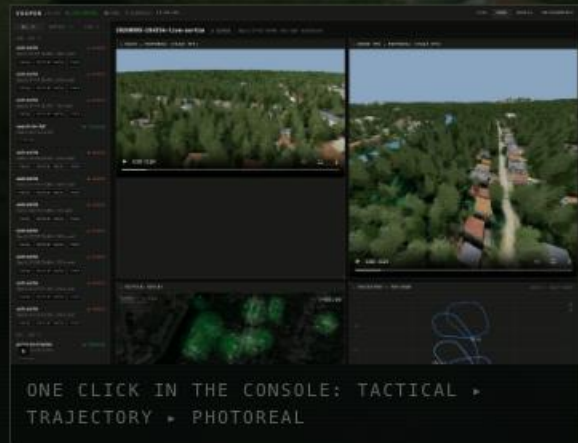
TERRAIN MESHING	DEM/DSM → collidable ground, filtered and true to relief
FOOTPRINT EXTRUSION	OSM tags → textured 3D buildings, physics included
PROCEDURAL CANOPY	60k solid trees a drone cannot cheat through — one fetch, endless world variants
RF SHADOW MODEL	baked jammer coverage raster — comms is terrain too
PPO • REINFORCEMENT LEARNING	reward-shaped search policy, millions of sim steps, on-device

— AFTER ACTION

Then fly it back — through the drone's eyes.

Every sortie replays photoreal in Isaac Sim — the same flight, frame for frame, from search to strike.

ISAAC RTX REPLAY OF THE RECORDED STRIKE SORTIE — SIMULATION RENDER, NOT COMBAT FOOTAGE.



ONE CLICK IN THE CONSOLE: TACTICAL ▶
TRAJECTORY ▶ PHOTOREAL

Point it at any coordinates **on Earth.**

Theo Oltean
tio3@cornell.edu

The cord is cut.